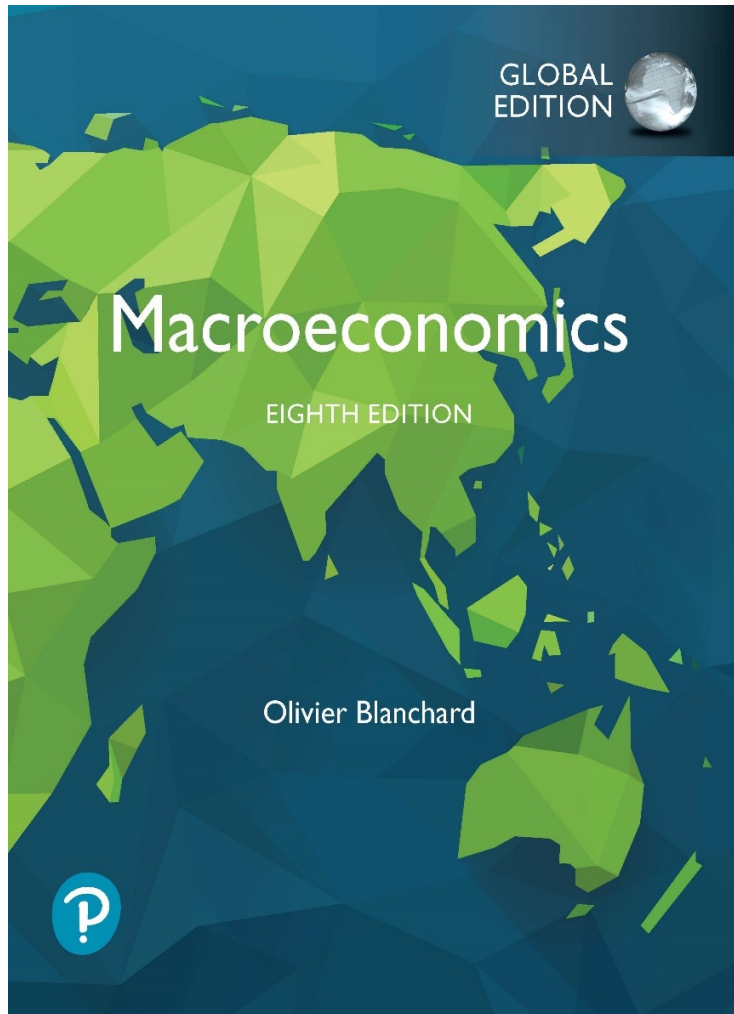


Macroeconomics

Eighth Edition, Global Edition



Chapter 3

The Goods Market

Chapter 3 Outline

The Goods Market

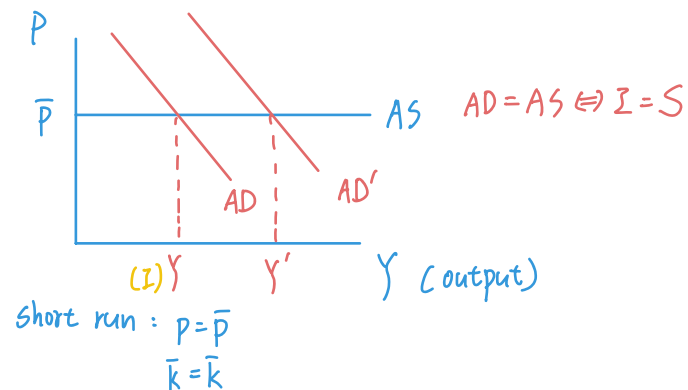
3.1 The Composition of GDP

3.2 The Demand for Goods

3.3 The Determination of Equilibrium Output

3.4 Investment Equals Saving: An Alternative Way of Thinking about Goods-Market Equilibrium

3.5 Is the Government Omnipotent? A Warning



The Goods Market

- When economists think about year-to-year movements in economic activity, they focus on the interactions among *demand*, *production*, and *income*:
 - Changes in the demand for goods lead to changes in production
 - Changes in production lead to changes in income
 - Changes in income lead to changes in the demand for goods

1. AS 為水平線 production 由 AD 決定

2. $Y = I(\text{income})$ change in AD $\rightarrow$ production, I

3. $D_x = D(P_x, P_y, \underset{\substack{\uparrow \\ \text{income}}}{I})$

Demand $\uparrow \rightarrow$ Production $\uparrow \rightarrow$ Income $\uparrow$


3.1 The Composition of GDP (1 of 3)

- **Consumption (C)**: goods and services purchased by consumers
- **Investment (I)** or *fixed investment*: the sum of nonresidential investment and residential investment
- **Government spending (G)**: purchases of goods and services by the federal, state, and local governments; excluding **government transfers**

(政府移轉)

$$AD: Y^D = C + I + G$$

↓
for long term use
(firms purchase)

3.1 The Composition of GDP (2 of 3)

- **Exports** (X): purchases of U.S. goods and services by foreigners
- **Imports** (IM): purchases of foreign goods and services by U.S. consumers, U.S. firms and the U.S. government
- **Net exports** or **trade balance**: $X - IM$
- Exports > Imports $\Leftrightarrow$ trade surplus (盈餘)
- Imports > Exports $\Leftrightarrow$ trade deficit (赤字)
- Inventory investment: difference between production and sales

$$AD: Y^D = C + I + G + (X - IM)$$

$\downarrow$
for long term use
(firms purchase)

$\rightarrow$ Net exports (NX)

銷售 > 生產 $\Rightarrow$ 存貨 $\downarrow$
< $\uparrow$

$$* \text{ Inventory Investment} = \text{Production} - \text{sales}$$

$$\text{Production} = \text{Inventory Investment} + \text{sales}$$

$$\text{Sales} = \text{Production} - \text{Inventory Investment}$$

3.1 The Composition of GDP (3 of 3)

Table 3.1 The Composition of U.S. GDP, 2018

		Billions of Dollars	Percent of GDP
	GDP (Y)	20,500	100.0
1	Consumption (C)	13,951	68.0
2	Investment (I)	3,595	17.5
	Nonresidential	2,800	13.6
	Residential	795	3.8
3	Government spending (G)	3,522	17.2
4	Net exports	-625	-3.0
	Exports (X)	2,550	12.4
	Imports (IM)	-3,156	-15.4
5	Inventory investment	56	0.2

(較高稅率)

國內民眾消費

trade deficit

順差 = 累積外匯存底 $X > IM$

Source: Survey of Current Business, February 2019, Table 1-1-5

3.2 The Demand for Goods (1 of 7)

$$Z \equiv C + I + G + X - IM$$

- The above identity defines the total demand for goods (Z) as consumption, plus investment, plus government, plus export, minus imports.
- In a closed economy ($X = IM = 0$):

$$Z \equiv C + I + G$$

3.2 The Demand for Goods (2 of 7)

- Consumption (C) is a function of disposable income (Y_D), which is the income that remains once consumers have received government transfers and paid their taxes.

$$C = C(Y_D) \quad (3.1) \quad \text{Demand for goods subject to } Y_D$$

(+)

- $C(Y_D)$ is called the **consumption function**.
- This is a **behavioral equation** that captures the behavior of consumers.

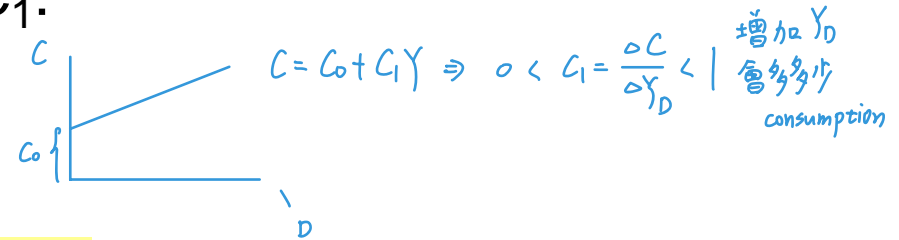
個人收入 (Y): disposable income $Y_D = Y - T$

最大化效用在有限的預算下 $\max U$
s.t. $P_X X + P_Y Y = M$

3.2 The Demand for Goods (3 of 7)

- Assume that the consumption function is a **linear relation** with two **parameters**, c_0 and c_1 :

$$C = c_0 + c_1 Y_D \quad (3.2)$$



- c_1 is the **propensity to consume**. 消費傾向 marginal propensity to consume
- c_0 is what people would consume if their disposable income equals zero.
- Changes in c_0 reflect changes in consumption for a given level of disposable income.

c_0 : sustainable living consumption 自發性消費

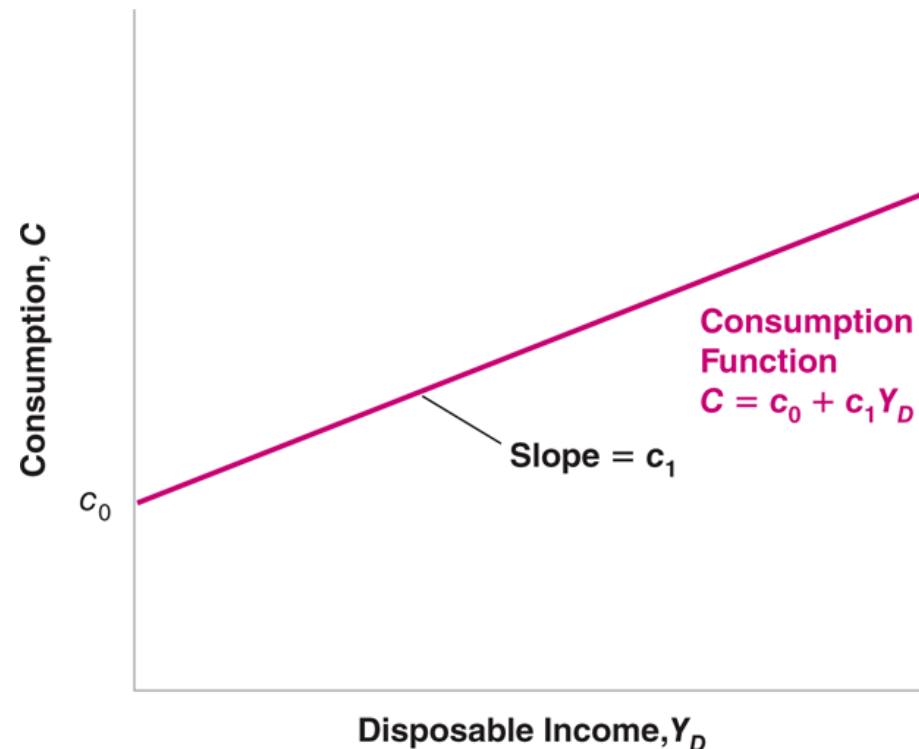
c_1 : 誘發性消費: 所得受影響

3.2 The Demand for Goods (4 of 7)

Figure 3.1 Consumption and Disposable Income

Consumption increases with disposable income but less than one for one.

A lower value of c_0 will shift the entire line down.



3.2 The Demand for Goods (5 of 7)

- Disposable income is:

$$Y_D \equiv Y - T$$

where Y is income and T is taxes minus government transfers.

- Replacing Y_D in equation (3.2) gives:

$$C = c_0 + c_1(Y - T) \quad (3.3)$$

3.2 The Demand for Goods (6 of 7)

- **Endogenous variables:** variables depend on other variables in the model (内生变量)
- **Exogenous variables:** variables not explained within the model but are instead taken as given (外生变量)

$$I = \bar{I} \quad (3.4)$$

- A bar on investment means investment is taken as given.

3.2 The Demand for Goods (7 of 7)

- T and G describe **fiscal policy**—the choice of taxes and spending by the government.
- G and T are exogenous because:
 - Governments do not behave with the same regularity as consumer or firms.
 - This book will typically treat G and T as variables chosen by the government and will not try to explain them within the model.

G, T, I

3.3 The Determination of Equilibrium Output (1 of 11)

- Assume $X=M=0$, so

$$Y^S = Y^D = Z \Rightarrow Y^*$$

$$Z \equiv C + I + G$$

- Replacing C and I from equations (3.3) and (3.4):

$$Z = c_0 + c_1(Y - T) + \bar{I} + G \quad (3.5)$$

- **Equilibrium in the goods markets** requires

$$Y = Z \quad (3.6)$$

- This is an **equilibrium condition**.

3.3 The Determination of Equilibrium Output (2 of 11)

- Replacing Z in (3.6) by equation (3.5) gives

$$\underset{(Y^*)}{Y} = c_0 + c_1(Y - T) + \bar{I} + G \quad (3.7)$$

- ✱ In equilibrium, *production (Y) is equal to demand, which in turn depends on income (Y), which is itself equal to production.*

3.3 The Determination of Equilibrium Output (3 of 11)

- Macroeconomists always use three tools:
 1. Algebra to make sure that the logic is correct
 2. Graphs to build the intuition
 3. Words to explain the results

3.3 The Determination of Equilibrium Output (4 of 11)

- Rewrite equation (3.7):

$$Y = c_0 + c_1 Y - c_1 T + \bar{I} + G$$

- Reorganize the equation:

$$(1 - c_1)Y = c_0 + \bar{I} + G - c_1 T$$

- Divide both sides by $(1 - c_1)$:

$$Y = \frac{1}{1 - c_1} [c_0 + \bar{I} + G - c_1 T] \quad (3.8)$$

multiplier effect: 增加 1% 的 G , 可以增加 $>1\%$ 的 Y
 外生變數決定 Y^*
 >1

which characterizes equilibrium output in algebra.

higher c_1 leads to higher Y^*
 ↪ propensity to consume
 c_0 : 自發性支出 (消費者信心)

$G - T > 0$ 預算赤字
 < 0 盈餘
 $= 0$ 平衡

3.3 The Determination of Equilibrium Output (5 of 11)

- **Autonomous spending:** $[c_0 + \bar{I} + G - c_1 T]$ 自發性支出
- Autonomous spending is positive because if $T = G$ (**balanced budget**) and c_1 is between 0 and 1, then $(G - c_1 T)$ is positive, and so is autonomous spending.
- The term $1/(1 - c_1)$ is the **multiplier**, which is larger when c_1 is closer to 1.
- If c_1 equals 0.6, the multiplier equals $1/(1 - 0.6) = 2.5$, meaning that an increase of consumption by \$1 billion will increase output by $2.5 \times \$1 \text{ billion} = \2.5 billion .

3.3 The Determination of Equilibrium Output (6 of 11)

- Steps to characterize the equilibrium graphically:
 1. Plot production as a function of income. Because production equals income, their relation is the 45-degree line.
 2. Plot demand as a function of income.

$$Z = (c_0 + \bar{I} + G - c_1T) + c_1Y \quad (3.9)$$

3. In equilibrium, production equals demand.



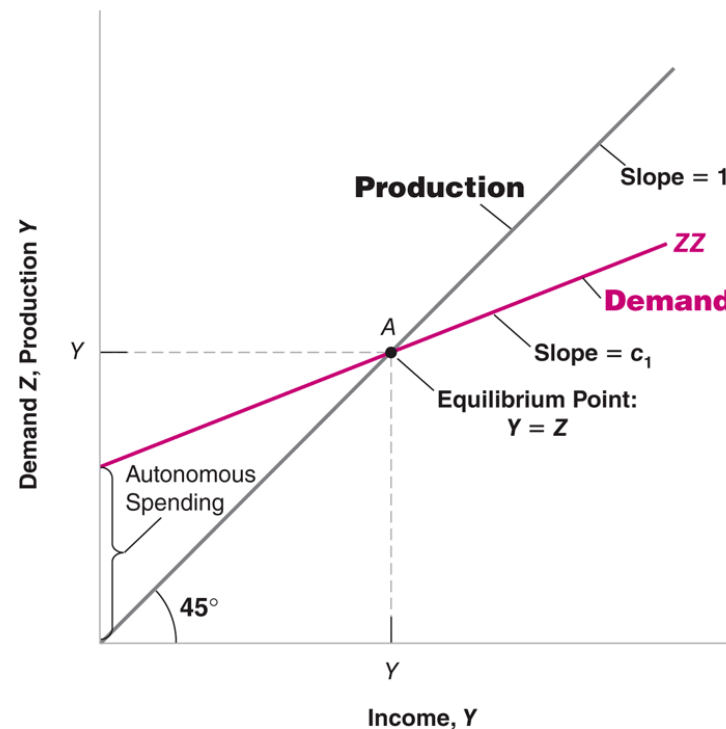
production = income

$$= c_0 + c_1(Y - T) + \bar{I} + G$$

3.3 The Determination of Equilibrium Output (7 of 11)

Figure 3.2 Equilibrium in the Goods Market

Equilibrium output is determined by the condition that production is equal to demand.



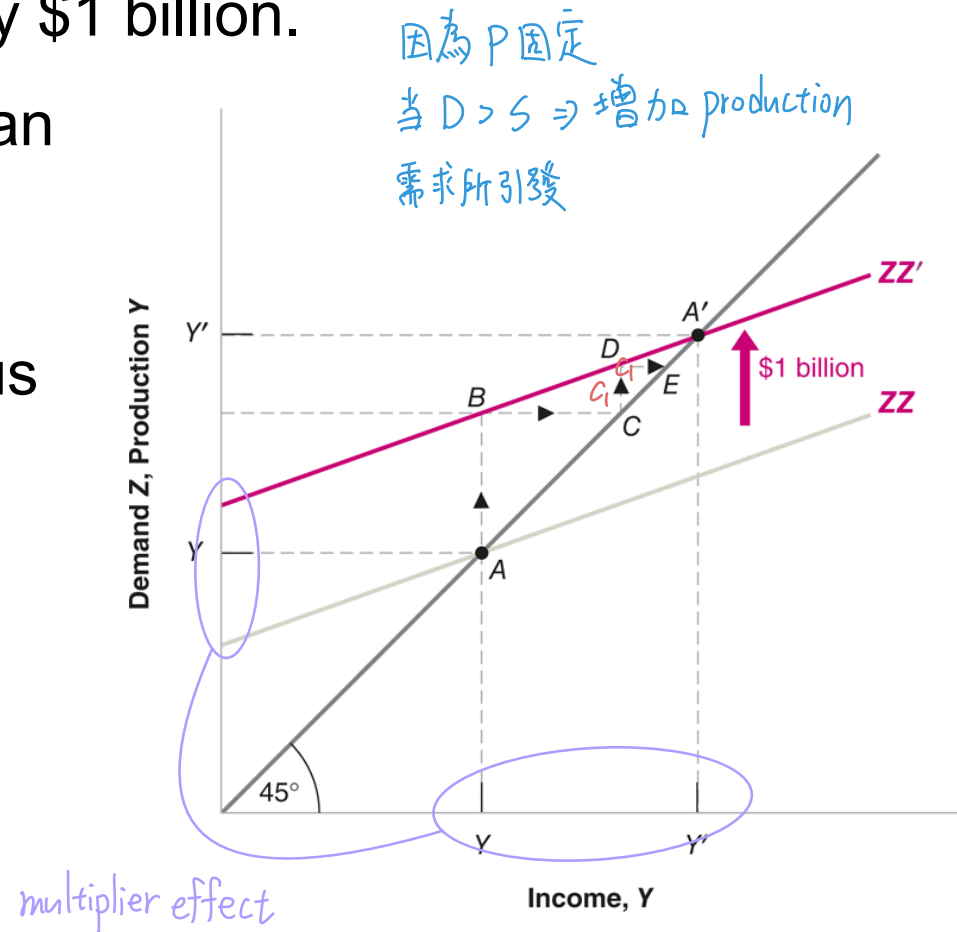
3.3 The Determination of Equilibrium Output (8 of 11)

- Suppose c_0 increases by \$1 billion.

Figure 3.3 The effects of an increase in autonomous spending on output

An increase in autonomous spending has a more than one-for-one effect on equilibrium output.

自發性消費改變 $c_0 \uparrow (I, G, T)$



3.3 The Determination of Equilibrium Output (9 of 11)

- *AB*: first-round increase in production
- *BC*: first-round increase in income
- *CD*: second-round increase in demand
- *DE*: second-round increase in production and income
- The total increase in production after $n+1$ rounds:

$$1 + c_1 + c_1^2 + \dots + c_1^n$$

which is a **geometric series** with a limit of $1/(1-c_1)$.

3.3 The Determination of Equilibrium Output (10 of 11)

- To summarize our findings using words:
 - Production depends on demand, which depends on income, which is itself equal to production.
 - An increase in demand leads to an increase in production and income, which in turn leads to a future increase in demand.
 - The ^{廠商的增長速度}increase in output that is larger than the initial shift in demand, by a factor equal to the multiplier.
 - The multiplier depends on the propensity to consume, which can be estimated using **econometrics**—the set of statistical methods used in economics.

3.3 The Determination of Equilibrium Output (11 of 11)

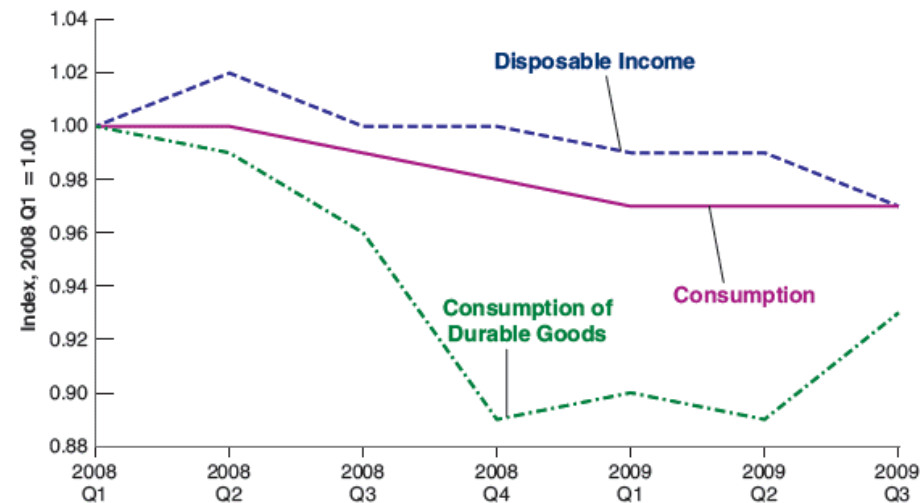
- The adjustment of output over time is called the **dynamics** of adjustment.
- How long the adjustment takes depends on how and when firms revise their production schedule.

FOCUS: The Lehman Bankruptcy, Fears of Another Great Depression, and Shifts in the Consumption Function (1 of 3)

- When people start worrying about the future, they decide to save more even if their current income has not changed.
- News about Lehman Brothers going bankrupt in September 2008 reminded people of the Great Depression, as confirmed by the number of searches for “Great Depression” in Google.
- Consumption fell even if disposable income had not yet changed.

FOCUS: The Lehman Bankruptcy, Fears of Another Great Depression, and Shifts in the Consumption Function (2 of 3)

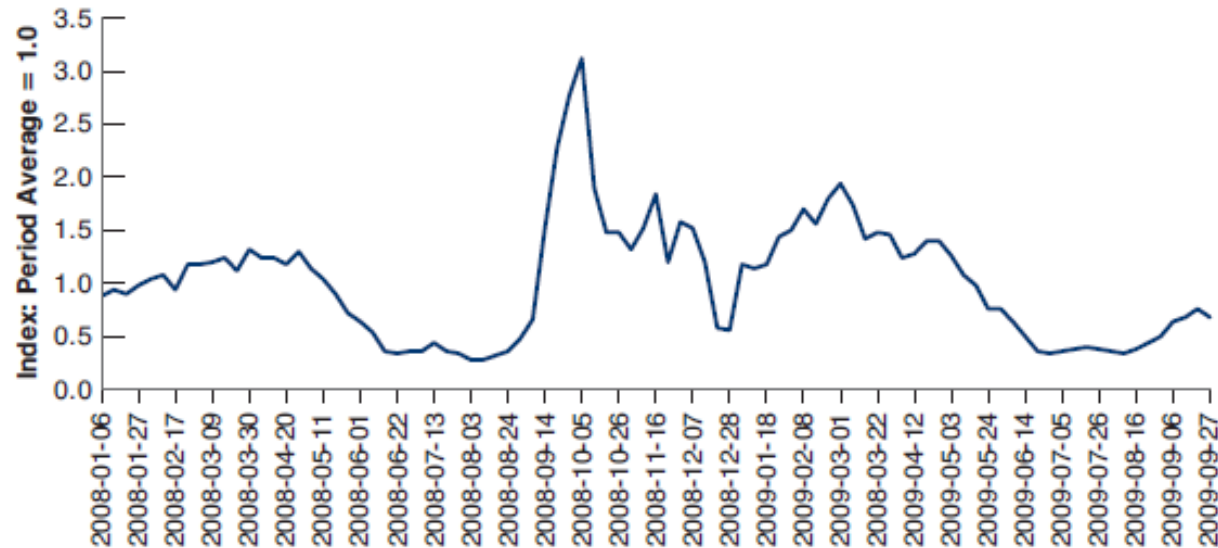
Figure 1 Disposable Income, Consumption, and Consumption of Durables in the United States, 2008:1 to 2009:3



Source: FRED: DPIC96, PCECC96, PCDGCC96.

FOCUS: The Lehman Bankruptcy, Fears of Another Great Depression, and Shifts in the Consumption Function (3 of 3)

Figure 2 Google Search Volume for “Great Depression,” January 2008 to September 2009



Source: Google Trends, “Great Depression.”

3.4 Investment Equals Saving: An Alternative Way of Thinking about Goods—Market Equilibrium (1 of 5)

$AD = AS$

$\Rightarrow Y = Y^D = Y^S$

- John Maynard Keynes articulated an alternative model that focuses instead on investment and saving in the General Theory of Employment, Interest and Money in 1936.
- **Private saving** (S) is

$$S \equiv Y_D - C$$
$$S \equiv Y - T - C$$

收到的稅收

- By definition, **public saving** = $T - G$. $\rightarrow$ 政府的支出
- Public saving $> 0 \Leftrightarrow$ **Budget surplus**
- Public saving $< 0 \Leftrightarrow$ **Budget deficit**

3.4 Investment Equals Saving: An Alternative Way of Thinking about Goods—Market Equilibrium (2 of 5)

- In equilibrium:

$$Y = C + I + G$$

- Subtract T from both sides and move C to the left side:

$$\underbrace{Y - T - C}_{\text{private saving}} = I + G - T$$

- The left side of the equation is simply S , so

$$S = I + G - T$$

- Or equivalently

$$I = S + (T - G) \quad (3.10) \quad \begin{array}{l} \text{private saving} \\ \text{public saving} \end{array}$$

- This is the **IS relation**, which stands for “Investment equals Saving”.

3.4 Investment Equals Saving: An Alternative Way of Thinking about Goods—Market Equilibrium (3 of 5)

- Two equivalent ways of stating the condition for equilibrium in the goods market:

Production = Demand

Investment = Saving

3.4 Investment Equals Saving: An Alternative Way of Thinking about Goods—Market Equilibrium (4 of 5)

- We can also derive equation (3.8) using equation (3.10).
- Because consumption behavior implies that:

$$S = Y - T - C$$
$$= Y - T - \underbrace{[c_0 - c_1(Y - T)]}_{\text{Consumption}}$$

$C = c_0 + c_1 Y_d$
↓
marginal propensity to consume

Rearranging terms, so

$$S = -c_0 + (1 - c_1)(Y - T) \quad (3.11) \quad Y = \left(\frac{1}{1 - c_1}\right) AE$$

- $(1 - c_1)$ is called the **propensity to save**, which is between zero and one.

3.4 Investment Equals Saving: An Alternative Way of Thinking about Goods—Market Equilibrium (5 of 5)

- In equilibrium, $I = S$, so that equation (3.10) becomes:

$$I = -c_0 + (1 - c_1)(Y - T) + (T - G)$$

$$I = S + (T - G)$$

- Solve for output:

$$Y = \frac{1}{1 - c_1} [c_0 + \bar{I} + G - c_1 T] \quad (3.12)$$

which is the same as equation (3.8).

FOCUS: The Paradox of Saving

- We are told about the virtues of thrift as we grow up, but the model in this chapter tells a different story.
- Suppose that consumers decide to save more, so c_0 decreases.
- Equation (3.12) implies that output decreases.
- Saving cannot change either, because equation (3.10) implies that at equilibrium:

$$I = S + (T - G)$$

- S cannot change because I , T or G does not change by assumption.

3.5 Is the Government Omnipotent? A Warning

- Equation (3.8) implies that the government can choose the level of G or T to affect the level of output it wants.
- However, there are many aspects of reality that we have not incorporated in our model:
 - Changing G or T is not easy.
 - Investment and imports may change, making it hard for governments to assess the effects of their policies (Chapters 5, 9, and 18 to 20).
 - Expectations are likely to matter (Chapters 14 to 16).
 - The effects on output may be unsustainable in the medium run (Chapter 9).
 - Cutting T or increasing G can lead to large budget deficits and public debt in the long run (Chapters 9, 11, 16 and 22).